

Indian Institute of Engineering Science and Technology, Shibpur
B. Tech (CST) 3rd Semester End-Semester Examination, November, 2024
Discrete Structures (CS2101)

Full Marks: 50

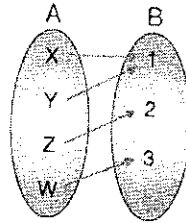
Time: 3 Hours

- Answer only five questions. No answer to extra question will be evaluated.
- Answer any three questions from Section-A and Section-B is mandatory
- Answer all parts to a same question together.
- Use diagram wherever possible

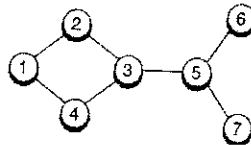
Section-A
(Answer any three questions)

1. Answer in one sentence only

(a) Which type of function is it?



- (b) If $A = \{x \in \mathbb{Z} \mid -6 \leq x \leq 6\}$, $B = \{x \in \mathbb{Z} \mid -3 \leq x < 3\}$ and p is the average of value in $A \cap B$, then $\lfloor p \rfloor = ?$
- (c) $(\exists i \in \{10, \dots, 30\}, i^2 < 2000) \wedge (i < 100) = ?$
- (d) What are the values of $m, n \in \mathbb{Z}$, where $(m, n) \in \{(x, y) \mid x^2 + y^2, x, y \in \mathbb{R}\}$
- (e) Height of binary tree is the number of edges between root and its furthest leaf. What is the height of a complete binary tree with 6 vertices (if possible)?
- (f) Conclude about the members in $(A - C) \cap (C - B)$ for $A = B \neq C$.
- (g) What is the chromatic number of the following graph?



- (h) If $NK(x)$ be the statement “ x has visited North Kolkata,” then what is the meaning of $\exists x \neg NK(x)$
- (i) Give a counter example of “If $a > b$ then $a^2 > b^2$, $a, b \in \mathbb{Z}$ ”.
- (j) If $x \vee (y \wedge z) = M \wedge (N \vee z)$, $x, y, z \in \{0, 1\}$ then what are the values of M and N ?

(1×10=10)

2. (a) Decide if $(B - A) \cup (C - A)$ is same as $(B \cup C) - A$ using set membership table.

(b) Let p and q be the propositions.

p : I bought a lottery ticket this week.

q : I won the million-dollar jackpot.

Express each of these propositions as an English sentence.

(i) $p \leftrightarrow q$ (ii) $p \wedge q$ (iii) $\neg p \rightarrow \neg q$ (i) $\neg p \wedge \neg q$ (i) $\neg p \vee (p \wedge q)$

(5 + 5 = 10)

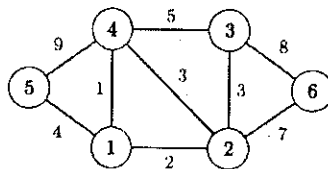
3. Prove the following as specified.

(a) $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n \geq 1$ (mathematical induction)

(b) $2\sqrt{k+1} + \frac{1}{\sqrt{k+1}} \geq 2\sqrt{k+2}$, $\forall k \in \mathbb{N}, k \geq 1$ (method of contradiction)

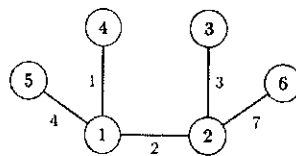
(5 + 5 = 10)

4. Consider a graph as follows.



(a) Represent the graph in form of both adjacency matrix and adjacency list. Hence compare two representations in terms of memory requirement.

(b) From a given graph, find a subgraph connecting all vertices without any cycle such that the sum of weights of all the edges is minimum. Show all the steps from the graph to the target subgraph.



(5 + 5 = 10)

5. (a) Solve the recurrence relation $F_n = 3F_{n-1} + 10F_{n-2} + 7.5^n$ where $F_0=4$ and $F_1=3$.

(b) If $1 + x + x^2 + x^3 + x^4 + \dots = \frac{1}{1-x}$, $|x| < 1$ then find G.F for the sequence $a_k = k + 1$,

$k = 0, 1, 2, 3, 4, \dots$

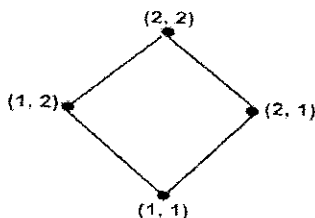
(5 + 5 = 10)

Section-B
(Mandatory)

6. Choose the correct alternative. No explanation is needed for your answer.

- (a) I: $R_1 = \{(0, 2), (1, 2), (2, 0)\}$ is not antisymmetric
 II: $R_2 = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2)\}$ is antisymmetric
 (i) I-True, II-False (ii) I-True, II-False (iii) I-True, II-True (iv) I-False, II-False

- (b) The following diagram represents (i) poset but not lattice (ii) both poset and lattice
 (iii) a relation but neither poset nor lattice (iv) not even a relation



- (c) What is the value of $S = 1 - p + p^2 - p^3 + \dots \infty$ for $p=3$?

(i) $1/4$ (ii) $-1/2$ (iii) $1/2$ (iv) Not computable

- (d) When $f(n) = n^3 + n \log n + 10$ then

(i) $f(n) = O(1)$ (ii) $f(n) = O(n)$ (iii) $f(n) = O(n^3)$ (iv) $f(n) = O(\log n)$

- (e) $\{x \mid x^3 - 3x^2 - x + 3 = 0\}$ is same as

(i) $\{t \mid t \in \mathbb{N}, t \leq 3\}$ (ii) $\{3, 1, -1\}$ (iii) $\{x \mid x^3 - 4x^2 + 3x = 0\}$ (iv) $\{3, 1, -1\}$ and $\{t \mid t \in \mathbb{N}, t \leq 3\}$

- (f) In mathematical induction, to prove $P(n)$ is true for all $n \geq 1$ which of the following steps is necessary?

(i) Prove $P(0)$ is true
 (ii) Assuming $P(K)$ is true, prove $P(K+1)$ is true
 (iii) Assuming $P(K+1)$ is false, prove a contradiction
 (iv) None of the above

- (g) What is the power set of the set $A = \{1, 2\}$?

(i) $\{\Phi, \{1\}, \{2\}, \{1, 2\}\}$ (ii) $\{\Phi, \{1, 2\}\}$ (iii) $\{\Phi, \{1\}, \{2\}\}$ (iv) $\{\{1, 2\}\}$

- (h) Which of the following is logically equivalent to the statement "If it rains, then the ground is wet"?

(i) The ground is wet if and only if it rains.
 (ii) If the ground is not wet, then it does not rain.
 (iii) If it does not rain, then the ground is not wet.
 (iv) If the ground is wet, then it rains.

- (i) Let $f: R \rightarrow R$ be a function such that $f(x+y) = f(x) \cdot f(y)$, $\forall x, y \in R$. If $f(1) = 2$ then what is $f(3)$?

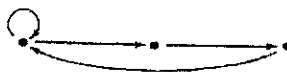
(i) 6 (ii) 8 (iii) 10 (iv) 18

- (j) In a binary tree, what is the maximum number of nodes at level l

(i) 2^l (ii) 2^{l-1} (iii) $2^l - 1$ (iv) 2^{l+1}

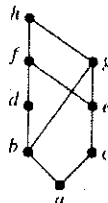
7. Clearly write **True** or **False** for the following statements. No justification is needed for your answer.

(a) The following digraph represents that relation is reflexive.



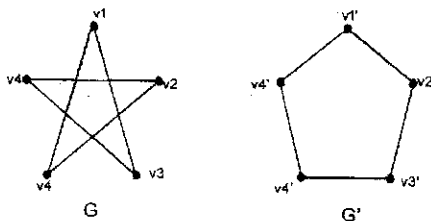
(b) $(A \cap B) \cup (\bar{A} \cap B) \cup (A \cap \bar{B}) \cup (\bar{A} \cap \bar{B})$ represents universal set.

(c) In the following diagram least upper bound of $\{b, c\}$ is $\{g, f, h\}$.



(d) $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are logically equivalent.

(e) G and G' are isomorphic graphs.



(f) In propositional logic, the expression $p \wedge (\neg p \vee q)$ is a tautology.

(g) A proof by contradiction assumes the negation of the statement to be proven and derives at a contradiction.

(h) G.F for the sequence $1, 1, 1, 1, \dots$ is $\frac{1}{1-x}$.

(i) A simple graph with n vertices can have at most n^2 edges.

(j) The composition of two transitive relations is always transitive.

(1×10 = 10)